

SHOHIN: Learned Temporal Ownership for Model-Owned Drafting, Revision, and Commitment

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Abstract

Can distinct learned owners for drafting, revision, and commitment convert an executed multi-stage inference budget into additional task accuracy? SHOHIN tests this question without a test-time external verifier or answer-level ensemble. On a protected 538-problem board, a dense Qwen3.5-9B draft→revision→whole-trajectory-commit system solves 383 problems versus 316 for a matched unchanged second pass. Trained revision also produces positive aggregate gains across five dense hosts spanning 0.8B–9B parameters and three families. Moving to sparse hosts, a 32,784-parameter incremental temporal gate reaches 143/256 on Qwen3.6-35B-A3B, versus 111 for a source-only host and 141 for the draft-matched trained-revision branch. A 1.66M-parameter GPT-OSS-120B revision reaches 111/256, versus 101 source-only and 103 self-refinement. On non-Qwen Mixtral-8x22B, a 3.54M-parameter revision surface scores 448/1,023 and beats draft-matched self-refinement by 92 answers, while exposing severe code and source-only-case retention regressions. The evidence supports parameter-efficient model-owned revision across three sparse families, but not a monotonic MoE scaling law or universal improvement: sparse commitment is untested, and the measured auxiliary selector is not conservative.

1 Introduction

Inference-time computation improves language-model performance when it is allocated through search, sampling, or adaptive refinement [10, 9]. These approaches raise a structural question: must extra compute be organized by an external verifier or answer aggregator, or can a model learn internally differentiated temporal roles that make later computation more capable than an unchanged second pass?

We study a simple hypothesis. A first model state should own a complete draft; a separately trained state should own full-trajectory revision; and a learned commitment mechanism should conservatively choose one coherent answer. This differs from asking the same model to “try again.” Prompted self-refinement can improve outputs [6], but intrinsic self-correction can also regress reasoning [5]. We therefore compare learned revision to both an unchanged second pass and matched self-refinement on the same identities. Dense comparisons match both controls on the model-owned draft and output budget; sparse comparisons use a source-only host as the unchanged baseline and self-refinement as the draft-matched control.

Our main result is a protected dense 9B release that solves 383/538 problems, compared with 316/538 for unchanged and 374/538 for trained revision alone. The effect is not isolated to one checkpoint: aggregate revision gains repeat across Qwen3.5-0.8B, 4B, and 9B, SmolLM3-3B, and OLMo2-7B. We then test whether the temporal principle transfers to sparse models without training their native routers or experts. A draft-conditioned gated system is strongest on a 35B-total Qwen MoE screen; trained post-MoE revision improves both controls on 117B-total GPT-OSS and produces a large replicated gain on 141B-total Mixtral.

The negative results are as important as the gains. Mixtral revision loses executable-code capability and retains only 64.6% of source-only-correct validation identities. An external selector restores code and 93.2% retention but gives back much of the gain; model-owned sparse commitment is untested. This separates capability acquisition from preservation of already-correct behavior.

Contributions.

1. A model-owned temporal architecture with separately learned draft, revision, and coherent commitment roles.
2. A protected 9B end-to-end capability result with immutable model, adapter, runtime, and result identities.
3. Aggregate revision transfer across five dense hosts and three model families.

Shohin assigns learned owners across inference time while freezing the host backbone

a Dense model-owned temporal lifecycle

b Sparse-host revision surface

c Tokenwise causal ownership

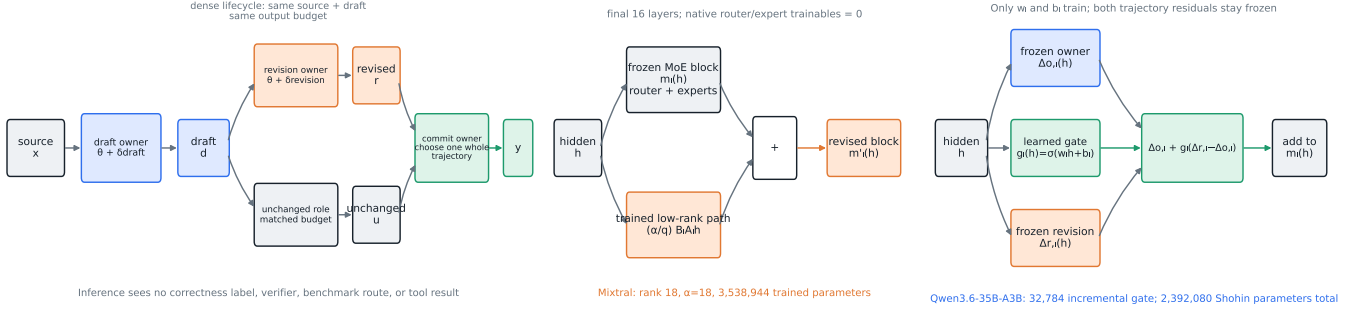


Figure 1: **Learned ownership across inference time.** (a) In the dense lifecycle, a model-owned draft feeds matched unchanged and trained-revision roles; commitment selects one whole trajectory. (b) The sparse-host revision surface augments the final frozen MoE blocks. (c) The causal gate interpolates frozen owner and revision residuals token by token.

4. Positive trained-revision gains across Qwen 35B, GPT-OSS 117B, and Mixtral 141B MoE screens, while keeping native sparse routers and experts frozen.
5. A measured capability–retention boundary that exposes the need for a future model-owned sparse commitment mechanism.

2 Architecture

Let $G(\theta, p; b)$ denote bounded generation by state θ from prompt p with output budget b . The dense lifecycle first produces

$$d = G(\theta + \delta_{\text{draft}}, x; b), \quad (1)$$

then a complete revision

$$r = G(\theta + \delta_{\text{revision}}, x || d; b). \quad (2)$$

In the dense lifecycle, a matched unchanged role produces a complete trajectory u from the same source and draft under the same budget. The commit owner emits one member of $\{u, r\}$; it cannot average logits or splice answer fragments. In the sparse experiments below, unchanged is instead a source-only host baseline; self-refinement is the draft-matched prompt control.

For a frozen feed-forward or MoE block m_l at layer l , the transferable Mixtral revision surface is

$$m'_l(h) = m_l(h) + \frac{\alpha}{q} B_l A_l h, \quad (3)$$

where q is residual rank. Only A_l and B_l are trained. The executed Mixtral transfer controls the final 16 layers, uses $q = 18$ and $\alpha = 18$, and trains 3,538,944 parameters. Native routers, experts, attention, embeddings, and the language-model head remain frozen.

The Qwen causal gate freezes learned owner and revision residual branches and trains only

$$g_l(h) = \sigma(w_l h + b_l), \quad (4)$$

$$m'_l(h) = m_l(h) + \Delta_{o,l}(h) + g_l(h) [\Delta_{r,l}(h) - \Delta_{o,l}(h)]. \quad (5)$$

Thus $g_l = 0$ recovers owner behavior and $g_l = 1$ recovers revision behavior. The reported gate adds 32,784 trainable parameters atop two frozen 1,179,648-parameter residual branches (2,392,080 SHOHIN parameters total) and uses response loss only, with no selector target or auxiliary routing label.

3 Experimental design

Every capability comparison holds evaluation identities and decoding fixed. Self-refinement receives the same draft as trained revision but uses a fixed natural-language correction instruction. Dense unchanged also receives that

draft; sparse unchanged is source-only. Thus sparse revision-versus-unchanged measures the full draft-conditioned pipeline against the host, while revision-versus-self-refinement isolates learned revision under matched draft access. Source-disjoint screens and holdouts are identity-separated from training inputs. Protected boards remain outside model-visible training and generation paths.

All gains are paired on exact identities. When available, we report discordant win/loss counts and exact two-sided McNemar tests. Parameter counts include only trained SHOHIN surfaces. Predeclared retention and per-domain gates remain binding even when aggregate accuracy improves, so a large numerical gain can still be a non-promotion.

The primary protected board contains GSM8K (100), MATH-500 (100), executable code (40), GPQA (198), and BBH logic (100), for 538 scored identities. The dense transfer boards contain 1,289 development or 1,279 holdout identities drawn from mathematics, logic/science, and execution-verified code. The MoE screens use a fixed source-disjoint 256-row board; Mixtral validation uses 1,023 independent rows (497 logic, 504 mathematics, 22 code).

4 Results

4.1 Protected dense release

Table 1: Protected Qwen3.5-9B result. Macro averages the five main domains; AIME is diagnostic and excluded from the 538 denominator.

System	Solved	Macro	Code	AIME
Unchanged second pass	316/538	67.263%	34/40	4/30
Trained revision	374/538	75.005%	35/40	3/30
Learned commit	383/538	75.815%	35/40	6/30
Coherent oracle (not deployable)	399/538	78.619%	35/40	6/30

The learned commit adds 67 solved problems over unchanged and nine over trained revision. An independently parameterized commit control solves 382/538, so the supported claim is useful learned coherent commitment, not a unique advantage of one antisymmetric scoring parameterization. Candidate-order consistency is 100% with zero swap error. Three prompt truncations in the product application are disclosed rather than silently recoded.

4.2 Dense transfer

Table 2: Trained revision versus matched unchanged. H and D denote holdout and development evidence.

Host	Split	Correct (revision / unchanged)	Delta
Qwen3.5-0.8B	H	328 / 242	+6.72 pp
Qwen3.5-4B	H	554 / 380	+13.60 pp
Qwen3.5-9B	H	625 / 495	+10.16 pp
SmolLM3-3B	D	469 / 358	+8.61 pp
OLMo2-7B	D	259 / 231	+2.17 pp

All five hosts improve in aggregate, but the table does not imply monotonic per-domain transfer. Qwen0.8B and SmolLM3 regress executable code; OLMo2 is too weak to promote; and the Qwen4 protected board fails its all-domain criterion.

4.3 Gated Qwen MoE transfer

On Qwen3.6-35B-A3B (35B total, 3B active), the temporal gate scores 143/256 versus 111/256 for the source-only host: +32 answers / +12.5 pp, with 38 paired wins and six losses ($p = 9.43 \times 10^{-7}$). This comparison measures the full draft-conditioned gated system, not the gate alone. Against the draft-matched trained-revision branch (141/256), the gate adds two answers (three wins, one loss; $p = .625$). It retains 105/111 source-only-correct identities. Mean revision weight rises from 0.643 at the first controlled layer to 0.903 at the last; these layer means describe the learned routing profile but do not by themselves exclude a fixed layer-specific gate.

4.4 GPT-OSS-120B transfer

Pinned GPT-OSS-120B has 117B total and 5.1B active parameters. A 1,658,880-parameter post-MLP revision surface controls the final 16 layers while native MXFP4 router and expert tensors remain frozen. On the same source-disjoint 256-row screen used for the other MoE points:

Table 3: GPT-OSS-120B matched screen.

Arm	Correct	Logic	Math	Code
Unchanged (source-only)	101	78	12	11
Self-refinement	103	81	11	11
Trained revision	111	82	18	11

Revision adds ten answers over source-only (19 wins / 9 losses, $p = .0872$) and eight over draft-matched self-refinement (12 / 4, $p = .0768$). Logic improves by four, mathematics by six, and all 11 MBPP solutions are preserved relative to source-only. The result nevertheless fails the frozen conservative boundary: only 92/101 (91.09%) source-only- correct identities remain correct. Mechanics, training, 12 independent one-H100 evaluation allocations, and CPU scoring complete with zero retries; charged usage is 26,322 H100-seconds (7.3117 H100-hours).

4.5 Cross-family Mixtral transfer

Mixtral-8x22B has 141B total and 39B active parameters. Its revision surface is 0.00251% of total and 0.00907% of active parameters. Both draft-conditioned arms use the same immutable Qwen3.6-35B-A3B drafts, making this a deliberately cross-family model-owned transfer.

Table 4: Mixtral validation on 1,023 identities.

Arm	Correct	Logic	Math	Code
Unchanged (source-only)	147	106	25	16
Self-refinement	356	224	121	11
Trained revision	448	272	176	0
Auxiliary selector	287	222	49	16

Revision gains 301 answers over the source-only host and 92 over the clean, draft-matched self-refinement control. The paired comparisons are 353 wins / 52 losses versus the source-only host ($p = 4.44 \times 10^{-56}$) and 131 / 39 versus self-refinement ($p = 7.92 \times 10^{-13}$). The 256-row screen shows the same ordering: 114 revision, 105 self-refinement, and 45 source-only.

The result is not a qualified universal release. Revision retains only 95/147 (64.6%) source-only-correct validation identities and loses all 22 code rows. A task-label-free three-way auxiliary linear selector restores 16/22 code and improves all domains over source-only, but scores only 287 and retains 137/147 (93.2%), three cases below the frozen 95% floor. It uses hashed source and completion features, including strong output-modality proxies, and is trained on per-arm correctness labels from the 256-row screen without task identity; it is not the in-model two-way commit used by the dense release. Its formal gate fails.

An identity-joined replay of all 1,023 scored candidate triples shows that the code failure is systematic (Table 5).

Table 5: Measured Mixtral output-form collapse. Boxed is over all 1,023 rows; the final three columns are over 22 MBPP rows.

Arm	Mean tokens	Boxed	Fences	Definitions	Code correct
Unchanged	436.13	203	22	21	16
Self-refinement	296.84	454	22	19	11
Trained revision	9.99	1,023	0	0	0

Revision’s median output is nine tokens overall and seven on MBPP; no code completion contains a return statement. The scorer requires explicit final answer form on answer-valued tasks, so the measured gain includes successful answer-format transfer and does not by itself isolate better reasoning. The surface violates executable output modality. The auxiliary selector restores code by selecting unmodified trajectories, localizing the next architecture problem to model-owned output-contract preservation. Without receiving a task label, the selector chooses unchanged

Shohin: temporal revision transfers; auxiliary selection exposes the retention boundary

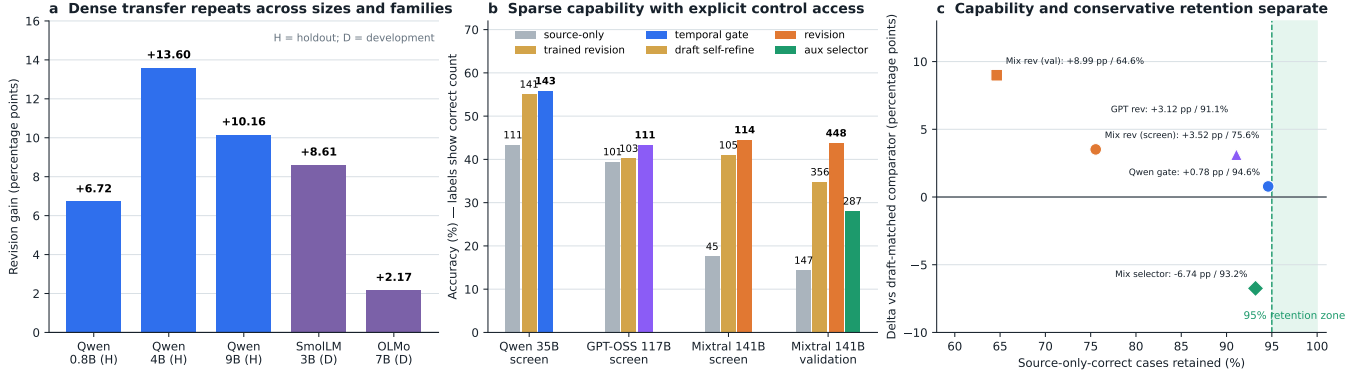


Figure 2: **Capability transfer and its retention boundary.** Dense gains repeat across sizes and families. Sparse bars label source-only and draft-conditioned arms explicitly. Retention is measured against source-only; capability delta is measured against each draft-matched comparator.

for all 22 MBPP identities and restores 16 correct programs. Yet it also selects unchanged for 460/504 mathematics identities and the stronger revision only 10 times. It can therefore recognize overt output modality, but not yet promote scored answer-valued gains while preserving that modality.

5 Discussion

Three observations survive the full evidence boundary. First, learned temporal revision consistently exceeds matched dense unchanged roles and beats both sparse controls on all three completed MoE-family screens. Sparse gains over source-only hosts additionally include access to the frozen draft. Second, the GPT-OSS and Mixtral gains require no trainable native routing or expert parameters; the Qwen gate is the strongest measured Qwen system but adds only two answers over its frozen trained-revision branch. Third, capability and retention are distinct optimization targets. Revision finds new correct trajectories but can overwrite already-correct behavior. The external auxiliary selector restores some conservatism but not the full large-host gain; model-owned sparse commitment has not yet been measured.

The three MoE screens do not support a monotonic scaling law. Qwen 35B, GPT-OSS 117B, and Mixtral 141B each show positive revision gains over both controls, but gain is nonmonotonic by active parameters, retention misses 95% at multiple points, and Mixtral code regresses. A weighted active-parameter slope is positive under an independent marginal sampling approximation, while the total-parameter interval crosses zero; three heterogeneous points are insufficient for a general scaling claim. Nemotron Super failed before science and Ultra has not run; neither is evidence.

6 Related work

Self-Refine [6] and Reflexion [8] use textual feedback or reflection. SHOHIN makes self-refinement a matched control and learns a separate revision role. LoRA [4] and ReFT [11] motivate parameter-efficient weight and representation interventions; our residual surfaces add an explicit temporal contract and are evaluated with model-owned prior trajectories.

Switch Transformers [1] route tokens among experts, while Mixture-of-Depths [7] routes tokens across layer compute. Recurrent-depth latent reasoning [2] and Coconut [3] extend internal computation without requiring every step to be natural language. SHOHIN tests a complementary axis: which learned temporal state should own a token or a complete answer after a draft already exists.

7 Limitations and reproducibility

The primary protected result is one pinned 9B host and should not be read as a universal theorem. Dense aggregate gains include domain failures. The Qwen MoE evidence is a 256-row development screen, and its source-only comparison does not isolate draft access from learned routing. GPT-OSS is also a 256-row screen; its positive paired differences

do not reach $p < .05$ and its retention gate fails. Mixtral validates at 1,023 rows but fails conservative retention and code; its scorer rewards required answer form, and no format-agnostic replay is claimed. The multi-stage systems also use more generation calls and tokens than a single unchanged pass, so these results are not compute-normalized. Public leaderboard superiority and a monotonic MoE scaling law are not supported.

All headline artifacts bind exact model revisions, trainable checkpoints, runtime manifests, source identities, arm outputs, and score reports. The protected 9B product report SHA-256 is `3e86751b...f7187`; the Qwen MoE screen is `1bd64551...b4871`; the GPT-OSS raw score is `906b3b01...d59`; the Mixtral validation raw score is `7befd864...c3d8`. Full hashes and the complete result ledger are in the companion evidence report. Both figures are deterministically generated by `pipeline/render_shohin_publication_figure.py`.

References

- [1] William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *arXiv preprint arXiv:2101.03961*, 2021.
- [2] Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. *arXiv preprint arXiv:2502.05171*, 2025.
- [3] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*, 2024.
- [4] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- [5] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations*, 2024.
- [6] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. Self-refine: Iterative refinement with self-feedback. *arXiv preprint arXiv:2303.17651*, 2023.
- [7] David Raposo, Sam Ritter, Blake Richards, Timothy Lillicrap, Peter Conway Humphreys, and Adam Santoro. Mixture-of-depths: Dynamically allocating compute in transformer-based language models. *arXiv preprint arXiv:2404.02258*, 2024.
- [8] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *arXiv preprint arXiv:2303.11366*, 2023.
- [9] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.
- [10] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.
- [11] Zhengxuan Wu, Aryaman Arora, Zheng Wang, Atticus Geiger, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. Ref: Representation finetuning for language models. *arXiv preprint arXiv:2404.03592*, 2024.